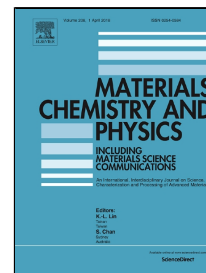


Accepted Manuscript

Size effects in physicochemical processes in nanoparticles and nanopores

V.V. Levdansky, O. Šolcová, P. Izák



PII: S0254-0584(18)30107-X

DOI: 10.1016/j.matchemphys.2018.02.014

Reference: MAC 20365

To appear in: *Materials Chemistry and Physics*

Received Date: 25 April 2017

Revised Date: 05 November 2017

Accepted Date: 14 February 2018

Please cite this article as: V.V. Levdansky, O. Šolcová, P. Izák, Size effects in physicochemical processes in nanoparticles and nanopores, *Materials Chemistry and Physics* (2018), doi: 10.1016/j.matchemphys.2018.02.014

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

SIZE EFFECTS IN PHYSICOCHEMICAL PROCESSES IN NANOPARTICLES AND NANOPORES

V.V. Levdansky*, O. Šolcová, P. Izák

Institute of Chemical Process Fundamentals of the CAS, v. v. i., Rozvojova 135, 165 02
Prague 6, Czech Republic

Highlights

Size effects in nanoparticles and confined ionic liquids are considered.

Lower cohesive energy in nanoparticles increases rate of chemical reaction.

Laplace pressure in nanoparticles decreases rate of chemical reaction.

Confined ionic liquids in nanopores affect gas mixture separation.

Keywords:

Size effects;

Nanoparticles;

Nanopores;

Diffusion.

Abstract

Mass transfer and chemical reactions in nanoscale particles under the assumption of the size dependence of the cohesive energy or the influence of the Laplace pressure are considered. A comparison of the above-mentioned approaches is given. It is shown that the use of the first or second of the above-mentioned approaches can lead to a difference in the dependence of the rates of the aforementioned processes on the nanoparticle size. The influence of the confinement of ionic liquids within nanopores of supported ionic liquid membranes on the separation of gas mixtures is discussed. Considered regularities can be

*Corresponding author.

E-mail address: vlev5@yahoo.com (V.V. Levdansky).

used for clarifying the physical mechanisms of the size effects, which take place in systems with nanoparticles and nanopores, and for the intensification of some processes in chemical engineering and nanotechnology.

1. Introduction

In recent years, physicochemical processes in systems with nanoparticles and nanopores have attracted the increasing attention of researchers in different branches of science and technology (e.g. chemical engineering, biotechnology, physics and chemistry of atmosphere, nanotechnology). It is known that many physicochemical processes that take place in nano-objects depend on the nano-object size. In particular, nano-effects leading to unusual properties of the electrospun nanofibers have been considered in [1, 2]. Nanomaterials have different properties from the bulk matter due to their high surface area to volume ratio and quantum effects in the nanoscale objects [1, 3]. One of the most important property from which we can derive many thermodynamic properties of materials is the cohesive energy [3]. The cohesive energy (that generally depends on the quantum effects) is responsible for the atomic structure, thermal stability, atomic diffusion, crystal growth and other properties [3–9]. As noted in [3], in modeling nanomaterials, two approaches are usually used. In the “top–down” approach, one looks at the variation of the properties of systems that change when going from the macro to the nano dimensions. The condition of the applicability of the aforementioned “top–down” approach (with respect to the minimum particle size) is analyzed in [3]. At the opposite, the “bottom–up” approach starts from atoms and more and more atoms are added, in order to see, how the properties are modified. The first from mentioned approaches makes use of classical physics, whereas the second relies on computational methods like molecular dynamics. Below we consider the size effects in nanoparticles on the basis of the aforementioned “top–down” approach and so-called “universal relation” which connects the cohesive energy with the activation energy for diffusion, the vacancy formation energy and the melting temperature [3, 7, 8]. As indicated in [3], the prediction from the aforementioned “universal relation” has been validated by comparison with available experimental results and existing theoretical models. This approach allows for the study of the size effects for different physicochemical processes in systems with nanoscale particles, in particular such as diffusion of atoms in nanoparticles [4], catalytic reactions on the nanoparticle surface [10], trapping of the impurity molecules by nanoparticles [11], physicochemical transformations in aerosol systems with nanoparticles [12]. It is worth

noting that the size dependence of the band gap of semiconductor nanoparticles is related to the dependence of the cohesive energy in nanoparticles on their size [13].

It is worth noting that in a number of papers the Laplace pressure concept is used to interpret the influence of the size of small spherical particles (including nanoscale particles) on physicochemical processes that take place in the particles. The Laplace pressure depends on the particle curvature and leads to the high pressure inside the nanoscale particles. This can influence the pressure-dependent processes taking place in nanoparticles. The Laplace pressure was used in particular in the study of processes such as diamond formation [14–16], nucleation in solutions in small droplets and pores [17], oxidation of Al nanoparticles [18]. It is worth noting that a number of authors discuss the correctness of using the notion of the curvature of the particle surface in studying nanoscale structures. As noted in [19], the surface curvature of the spherical nanoparticle is actually only a convenient parameter that allows us to describe the ratio of the surface area to the volume of the particle and has no relation to physics of the size effects. The author of [20] notes that the phrase often used in papers “phenomena due to a high curvature” should be replaced by the phrase “phenomena due to a high specific surface area”. It is worth noting that physicochemical properties of the flat nanoscale thin film depend on the film thickness [21, 22] despite the fact that in the present case the curvature and accordingly the Laplace pressure are equal to zero. The discussion of the applicability of the Laplace pressure concept to nanoparticles is given in [22–25].

The approaches related to the size dependence of the cohesive energy (and accordingly the activation energy for diffusion) in nanoparticles and the Laplace pressure concept may lead to the different (opposite) dependence of the rate of some physicochemical processes in nanoparticles on the nanoparticle size. This fact can be used for clarifying the physical mechanisms of size effects in nanoparticles.

It is worth noting that the confinement of ionic liquids in nanopores can also affect the physicochemical processes occurring in nanopores. For example, the confinement of ionic liquids in nanopores can affect mass transfer in nanopores and the separation of gas mixtures at their diffusion through nanopores of supported ionic liquid membranes. At first, let us consider the influence of the size effects on mass transfer and chemical reactions in nanoparticles.

2. Model

2.1. Effect of the size-dependent cohesive energy and accordingly the activation energy for diffusion on mass transfer in nanoparticles

It should be noted that many physicochemical processes in nanoparticles depend on mass transfer in them. The characteristic time for mass transfer by diffusion of atoms (molecules) in the macroscopic spherical aerosol particle $t_{m\infty}$ can be written as [26]

$$t_{m\infty} = \frac{d^2}{4\pi^2 D_{\infty}} \quad (1)$$

where d is the particle diameter, D_{∞} is the diffusion coefficient of atoms in the macroscopic spherical aerosol particle (the value of D_{∞} does not depend on the particle size).

The characteristic time for mass transfer by diffusion of atoms in the spherical aerosol nanoparticle t_{mp} under the use of the aforementioned “top-down” approach can be written in a form similar to Eq. (1) with the replace of D_{∞} by the size-dependent diffusion coefficient for the nanoparticle D_p .

Let us consider the dimensionless characteristic time for mass transfer in the nanoparticle t_{mp}^*

$$t_{mp}^* = \frac{t_{mp}}{t_{m\infty}} = \frac{D_{\infty}}{D_p} \quad (2)$$

The diffusion coefficient of atoms in the nanoparticle is given by [4]

$$D_p = D_0 \exp\left(-\frac{E_{dp}}{RT}\right) \quad (3)$$

where D_0 is the pre-exponential factor (it is assumed for simplicity to be constant and equal to the value for bulk matter), E_{dp} is the activation energy for diffusion in the nanoparticle, R is the gas constant, T is the temperature.

It is known that the activation energy for diffusion in nanoparticles is size-dependent [4]. The size dependence of the diffusion coefficient in nanoparticles is associated with the dependence of the cohesive energy and accordingly the activation energy for diffusion in the nanoparticle on the nanoparticle size. The values of the cohesive energy E_{cp} , the activation energy for diffusion E_{dp} , the activation energy for vacancy formation E_{vp} and the melting temperature T_{mp} for the nanoparticle can be related to the same values for bulk matter ($E_{c\infty}, E_{d\infty}, E_{v\infty}, T_{m\infty}$) by the following “universal relation” [3, 7, 8]:

$$\frac{E_{cp}}{E_{c\infty}} = \frac{E_{dp}}{E_{d\infty}} = \frac{E_{vp}}{E_{v\infty}} = \frac{T_{mp}}{T_{m\infty}} \quad (4)$$

The melting temperature of the free metallic, organic and semiconductor nanoparticles usually decreases with a decrease in the nanoparticle size [6]. The melting temperature of nanoparticles embedded in the surrounding matrix depends on the physicochemical properties of the matrix and can be both higher and lower than the melting temperature of the bulk material [27]. It is worth noting that nanoparticles embedded in the surrounded matrix are applied in modern chemical engineering and biotechnology. For example, magnetic nanoparticles are used in [28] to improve the separation of gas mixtures by polymeric membranes. In the case of the supported ionic liquid membranes, which are used in particular for the gas mixture separation, the confined ionic liquid in the nanopore of the membrane can be considered as the embedded nano-object in the membrane. At first, let us consider the case of free nanoparticles. There are many models describing the size dependence of the nanoparticle melting temperature. In particular, according to [29] the size dependence of the melting temperature of the free nanoparticle can be presented in the form

$$T_{mp} = T_{m\infty} \exp\left(-\frac{4\delta}{\delta + d}\right) \quad (5)$$

where d is the nanoparticle diameter, δ is the Tolman length. In the case when $4\delta/d \ll 1$ Eq. (5) is reduced to

$$T_{mp} = T_{m\infty} \left(1 - \frac{4\delta}{d}\right) \quad (6)$$

The size dependence of the melting temperature for the free nanoparticle similar to Eq. (6) is given in a number of papers in particular in [6, 30]. The value of E_{dp} and the diffusion coefficient of atoms in the nanoparticle D_p in view of Eqs. (3), (4) and (6) can be written as

$$E_{dp} = E_{d\infty} \left(1 - \frac{4\delta}{d} \right) \quad (7)$$

$$D_p = D_0 \exp \left(-\frac{E_{d\infty}}{RT} + \frac{4E_{d\infty}\delta}{dRT} \right) = D_\infty \exp \left(\frac{\alpha_1}{d} \right) \quad (8)$$

where $\alpha_1 = 4E_{d\infty}\delta/(RT)$, $D_\infty = D_0 \exp\{-E_{d\infty}/(RT)\}$.

It can be seen from Eq. (8) that when the value of $4\delta/d$ tends to zero, the value of D_p tends to D_∞ .

2.2. Laplace pressure and mass transfer in small spherical particles

Let us consider the possible influence of the Laplace pressure ΔP_L on the physicochemical processes in the small spherical particle. The value of ΔP_L is determined from the Laplace-Young equation [14, 15]

$$\Delta P_L = P_{in} - P_{out} = \frac{4\sigma}{d} \quad (9)$$

where P_{in} is the pressure inside the particle, P_{out} is the pressure outside the particle, σ is the surface energy density.

It should be noted that Eq. (9) is widely used for bubbles in bubble electrospinning or bubbl spinning for mass-production of nanofibers. As noted in [2], this equation is still valid for nanobubbles.

The diffusion coefficient in the condensed phase generally depends on the pressure in the substance [31]. With the use of the Laplace pressure concept, in the case when $P_{out} \ll$

$4\sigma/d$, taking into account [31], we can write the following equation for the diffusion coefficient of atoms (molecules) in the small **spherical** particle D_{pL} :

$$D_{pL} = D_0 \exp\left(-\frac{E_{d\infty}}{RT} - \frac{4\sigma\Delta V}{dRT}\right) = D_\infty \exp\left(-\frac{\alpha_2}{d}\right) \quad (10)$$

where ΔV is the activation volume for diffusion, $\alpha_2 = 4\sigma\Delta V/(RT)$.

The characteristic time for mass transfer by diffusion of atoms in the small spherical aerosol particle t_{mp} in the case of using the Laplace pressure concept can be written in a form similar to Eq. (1) with the replace of D_∞ by the diffusion coefficient D_{pL} .

Assuming that the above-mentioned size effects associated with the size dependence of the cohesive energy (and accordingly the activation energy for diffusion) or with the Laplace pressure can be jointly realized in a nanoscale particle, we can write the following equation for the effective value of the activation energy for diffusion E_{eff} :

$$E_{eff} = E_{d\infty} - \Delta E_{d1} + \Delta E_{d2} \quad (11)$$

where the value of ΔE_{d1} characterizes a decrease in the value of E_{eff} with reduction in the particle size due to the size dependence of the cohesive energy and accordingly the activation energy for diffusion, the value of ΔE_{d2} is related to an increase in E_{eff} with reduction in the particle size due to the effect of the Laplace pressure. In view of the aforementioned, the values of ΔE_{d1} and ΔE_{d2} can be written as

$$\Delta E_{d1} = \frac{4E_{d\infty}\delta}{d} \quad (12)$$

$$\Delta E_{d2} = \frac{4\sigma\Delta V}{d} \quad (13)$$

The effective diffusion coefficient in the nanoparticle D_{eff} under assumption of the joint influence of the aforementioned size effects with considering Eqs. (11)–(13) is given by

$$D_{\text{eff}} = D_{\infty} \exp\left(\frac{\alpha_3}{d}\right) \quad (14)$$

where $\alpha_3 = 4(E_{\text{dso}}\delta - \sigma\Delta V)/(RT)$.

It is seen from Eq. (14) that in the case when $E_{\text{dso}}\delta/(\sigma\Delta V) > 1$ the value of D_{eff} increases with a reduction in the particle size (for simplicity, we assume the values of δ , σ and ΔV to be constant and equal to the values for bulk matter).

3. Results and discussion

3.1. Influence of size effects on the characteristic time of mass transfer in nanoparticles

Let us consider the dependence of the dimensionless characteristic time of mass transfer t_{mp}^* on the particle size at the diffusion of the copper-64 tracer in pure copper [31] in the case of the spherical copper nanoparticle. Figure 1 shows the dependence of the value of t_{mp}^* on the diameter of the nanoparticle for the aforementioned approaches to the description of the size effects in the nanoparticle at $T = 1000$ K, $\sigma = 1.83$ J/m² [32], $\delta = 0.26$ nm. In view of [31], the values of ΔV and E_{dso} were taken as equal to $6.25 \cdot 10^{-6}$ m³/mol and 196 kJ/mol respectively.

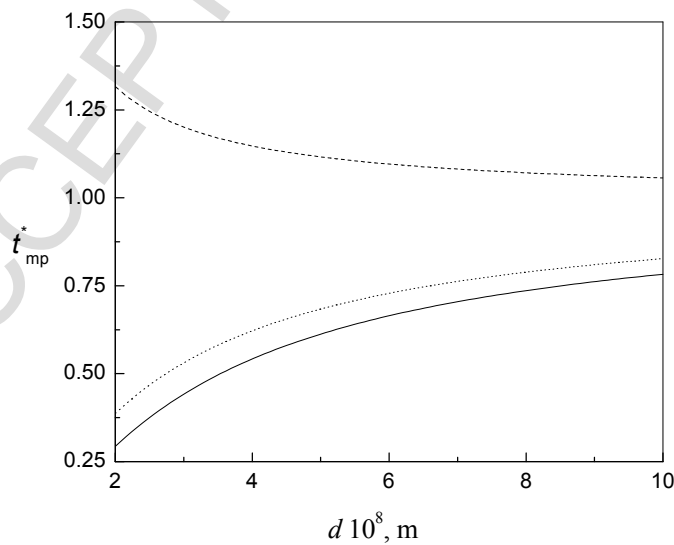


Fig. 1. Dependence of t_{mp}^* on the nanoparticle diameter d in diffusion of the copper-64 tracer in pure copper; solid line: the effect of the size-dependence of the cohesive energy and accordingly the activation energy for diffusion, dashed line: the Laplace pressure effect, dotted line: the joint influence of the mentioned effects.

Figure 1 shows that the value of t_{mp}^* is lower in the use of the approach related to the size dependence of the cohesive energy than in the use of the Laplace pressure concept. A reduction in the nanoparticle size leads to a decrease in t_{mp}^* by using the first of the aforementioned approaches and to an increase in t_{mp}^* by using the second approach. The dependence of t_{mp}^* on the nanoparticle size obtained under the assumption of the joint influence of the aforementioned size effects is similar to the result obtained by taking into account only the size dependence of the cohesive energy and accordingly the activation energy for diffusion. It is worth noting that the experimental data usually show an increase in the diffusion coefficient in the nanoparticle with a decrease in its size [33] that leads to a decrease in the characteristic time for mass transfer.

3.2. Size dependence of rate constants of chemical reactions in nanoparticles

Chemical reactions in nanoparticles are of interest for a number of branches of chemical engineering and nanotechnology as well as for processes that take place in the atmosphere. For example, it is known that chemical reactions in aerosol particles cause their aging [26]. Let us consider the bimolecular chemical reaction inside the spherical aerosol nanoparticle. The rate of the chemical reaction j in view of [34] can be written as

$$j = k_{\text{eff}} n_{\text{Ap}} n_{\text{Bp}} \quad (15)$$

where n_{Ap} and n_{Bp} are the concentrations of the reactants A and B in the nanoparticle, which are assumed for simplicity to be uniform by particle volume, k_{eff} is the effective rate constant for the bimolecular chemical reaction in the nanoparticle that is given by

$$k_{\text{eff}} = \frac{k_{\text{dp}} k_{0\text{p}}}{k_{\text{dp}} + k_{0\text{p}}} \quad (16)$$

Here $k_{0\text{p}}$ is the intrinsic rate constant of the bimolecular chemical reaction in the nanoparticle, k_{dp} is the rate constant of the chemical reaction in the nanoparticle in the diffusion-controlled regime of the reaction. In the diffusion-controlled regime of the chemical reaction the value of k_{eff} can be written as [34, 35]

$$k_{\text{eff}} = k_{\text{dp}} = 4\pi r_0 (D_{\text{Ap}} + D_{\text{Bp}}) \quad (17)$$

where D_{Ap} and D_{Bp} are the diffusion coefficients for the reactants A and B in the nanoparticle, r_0 is the encounter radius at which the reaction can occur (it is usually assumed that for the bimolecular reaction $r_0 = r_{\text{A}} + r_{\text{B}}$, where r_{A} and r_{B} are the radii of the molecules of the components A and B [35]).

It is worth noting that, as indicated in [26], the chemical aging of the semisolid organic aerosol particles is likely to be limited by the rate of diffusion of the reactants in the particles.

In the use of the first of the above-mentioned approaches (related to the size dependence of the cohesive energy and accordingly the activation energy for diffusion) the value of the rate constant of the chemical reaction in the spherical nanoparticle in the diffusion-controlled regime in view of Eqs. (8) and (17) can be written as

$$k_{\text{dp}} = k_{\text{dp1}} = 4\pi r_0 \left[D_{\text{A}\infty} \exp\left(\frac{\alpha_{1\text{A}}}{d}\right) + D_{\text{B}\infty} \exp\left(\frac{\alpha_{1\text{B}}}{d}\right) \right] \quad (18)$$

where $\alpha_{1\text{A}} = 4E_{\text{dA}\infty}\delta/(RT)$, $\alpha_{1\text{B}} = 4E_{\text{dB}\infty}\delta/(RT)$ and $D_{\text{A}\infty}$, $D_{\text{B}\infty}$, $E_{\text{Ad}\infty}$, $E_{\text{dB}\infty}$ are respectively the diffusion coefficients and activation energies for diffusion of the reactants A and B in bulk matter.

In the use of the second approach (related to the Laplace pressure concept) the rate constant of the chemical reaction in the spherical nanoparticle in the diffusion-controlled regime in view of Eqs. (10) and (17) is given by

$$k_{dp} = k_{dp2} = 4\pi r_0 \left[D_{A\infty} \exp\left(-\frac{\alpha_{2A}}{d}\right) + D_{B\infty} \exp\left(-\frac{\alpha_{2B}}{d}\right) \right] \quad (19)$$

where $\alpha_{2A} = 4\sigma\Delta V_A / (RT)$, $\alpha_{2B} = 4\sigma\Delta V_B / (RT)$ and ΔV_A , ΔV_B are the activation volumes for diffusion of the reactants A and B.

Equations (18) and (19) show that $k_{dp1} > k_{dp2}$ and k_{dp1} increases and k_{dp2} decreases with a decrease in the nanoparticle size.

In the case of the recombination of atoms in the spherical nanoparticle, the rate constant of the recombination reaction k'_{dp} in the diffusion-controlled regime in view of [36] is given by

$$k'_{dp} = 4\pi r_0 D'_p \quad (20)$$

where D'_p is the diffusion coefficient of atoms participating in the recombination reaction in the nanoparticle.

The dimensionless value of the rate constant of the recombination reaction in the nanoparticle can be written as

$$k_{dp}^* = \frac{k'_{dp}}{k'_{d\infty}} \quad (21)$$

where $k'_{d\infty}$ is the rate constant of the recombination reaction in bulk matter in the diffusion-controlled regime of the reaction ($k'_{d\infty} = 4\pi r_0 D'_\infty$, where D'_∞ is the diffusion coefficient of the atoms participating in the recombination reaction in bulk matter).

In the case when the size effect is related to the size dependence of the cohesive energy in the nanoparticle (and accordingly the size dependence of the activation energy for diffusion), in view of Eq. (8) we have the following equation for the dimensionless rate constant of the recombination reaction:

$$k_{dp1}^* = \exp\left(\frac{\alpha'_1}{d}\right) \quad (22)$$

where $\alpha'_1 = 4E'_{d\infty}\delta/(RT)$, $E'_{d\infty}$ is the activation energy for the diffusion of atoms participating in the recombination reaction without consideration of the size effect.

In the case when the size effect is related to the Laplace pressure in the spherical nanoparticle, in view of Eq. (10) we can write the following equation for the dimensionless rate constant of the recombination reaction:

$$k_{dp2}^* = \exp\left(-\frac{\alpha'_2}{d}\right) \quad (23)$$

where $\alpha'_2 = 4\sigma\Delta V'(RT)$, $\Delta V'$ is the activation volume for the diffusion of atoms participating in the recombination reaction).

Taking into account Eqs. (2), (20) and (21), we have the relation $k_{dp}^* = 1/t_{mp}^*$. Figure 2 shows the model calculations of the size dependence for k_{dp1}^* and k_{dp2}^* at different values of the parameters α'_1 and α'_2 .

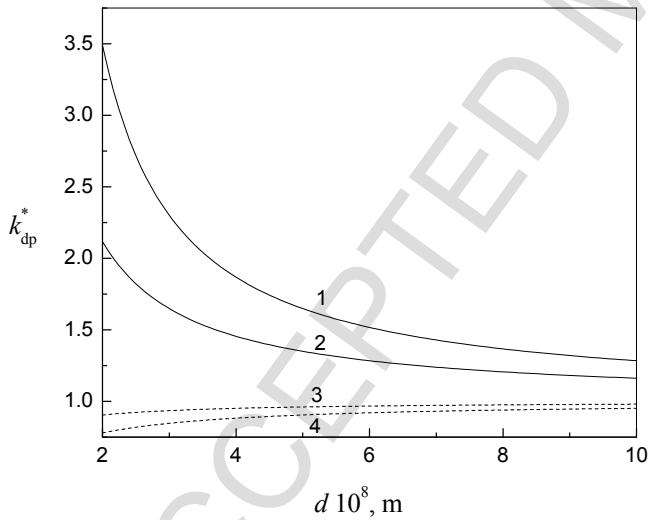


Fig. 2. Dependence of k_{dp}^* on the nanoparticle diameter d ; solid lines: the effect of the size dependence of the cohesive energy and accordingly the activation energy for diffusion ($k_{dp}^* = k_{dp1}^*$), dashed lines: the Laplace pressure effect ($k_{dp}^* = k_{dp2}^*$); 1: $\alpha'_1 = 2.5$; 2: $\alpha'_1 = 1.5$; 3: $\alpha'_2 = 0.2$; 4: $\alpha'_2 = 0.5$.

Equations (22), (23) and Fig. 2 show that $k_{dp1}^* > k_{dp2}^*$ and that k_{dp1}^* increases and k_{dp2}^* decreases with a decrease in the nanoparticle size.

3.3. Effect of confined ionic liquids in nanopores on separation of gas mixtures

It is worth noting that the size effects similar to the aforementioned effects for nanoparticles can take place also in the problems related to ionic liquids confined within nanoscale pores.

The confinement of ionic liquids in nanoscale pores can affect the critical temperature of solid-to-liquid phase transitions (the melting point) and the diffusivity of molecules in ionic liquids [37–39]. The melting point for confined ionic liquids is usually lower and the diffusivity of molecules is higher than for unconfined ionic liquids. The similar regularities for the aforementioned values take place for the above-considered case of nanoparticles at the size-dependent cohesive energy. A decrease in the melting point of confined liquids (including ionic confined liquids) and an increase in the diffusivity of molecules in them can be explained by a lower coordination number in confined liquids [39, 40]. It is worth noting that the coordination number can be related to the cohesive energy (a decrease in the coordination number leads to a decrease in the cohesive energy) [41]. As indicated in [40], the mobility of solvent molecules greatly affects the reactivity and dynamics of solutes. Thus, we can expect that the rate constants of chemical reactions in liquids confined in nanopores (including the confined ionic liquids) can increase with reduction in the nanopore size (similarly to the above-considered case of an increase in the rate constants of chemical reactions in nanoparticles at the size-dependent cohesive energy and accordingly the activation energy for diffusion). The solubility in confined liquids (including ionic confined liquids) can be higher than in unconfined liquids [38, 42]. The method of molecular dynamics simulation is usually used in a theoretical study of mass transfer in confined ionic liquids. The confinement of ionic liquids can cause specific features of the course of the physicochemical processes in confined ionic liquids. It is worth noting that atomistic simulations of the ionic liquid $C_6C_{1im}Tf_2N$ confined within carbon nanotubes (CNTs) indicate highly ordered structures within the CNT, with cations segregated at the tube wall and anions at the tube center [38]. As indicated by authors of [38], this nanoscale organization can result in an increase in the ionic liquid self-diffusivity as well as the solubility and the diffusivity of molecules such as CO_2 and H_2 . The confinement of the aforementioned ionic liquid in the CNT can also increase the selectivity for the separation of CO_2/H_2 mixture [38].

The ideal selectivity in the separation of gas mixtures in the diffusion of molecules through the supported ionic liquid membranes with nanopores filled with the ionic liquid according to the solution-diffusion model [42] can be written as

$$\alpha_{ij} = \frac{D_i(r)S_i(r)}{D_j(r)S_j(r)} \quad (24)$$

where r is the mean nanopore radius, D_i and D_j are diffusion coefficients for the components i and j of the mixture in the confined ionic liquid, S_i and S_j are solubilities for the components i and j of the mixture in the confined ionic liquid.

In the case when the value of $D_i S_i$ increases with reduction in r faster than the value of $D_j S_j$, the ideal selectivity α_{ij} will increase with a decrease in r . Thus, the confinement of the ionic liquids in nanopores of the supported ionic liquid membranes can be used to control mass transfer of dissolved molecules of the components i and j in nanopores and accordingly to manage the separation of the gas mixture.

4. Conclusions

Two approaches to the description of the influence of size effects on mass transfer and chemical reactions in nanoparticles are considered. The first one is related to the size dependence of the cohesive energy and accordingly the activation energy for diffusion of atoms (molecules) in nanoparticles. The second one is related to the high curvature of nanoparticles and accordingly the high Laplace pressure in them. It is shown that the use of the first or second of the aforementioned approaches can lead to the different (opposite) size dependence of the characteristic time of mass transfer as well as the rate constant of the chemical reaction. The size effects in the supported ionic liquid membranes are discussed. The confinement of the ionic liquids in nanopores of the membranes can be used in particular to control diffusion of dissolved gas molecules through the supported ionic liquid membranes with nanopores and accordingly to manage the membrane separation of gas mixtures (e.g. the CO_2/H_2 mixture). The results of the paper can be used for clarifying the physical mechanisms of the size effects in nano-objects and for a more correct description of some processes that take place in various fields of chemical engineering and nanotechnology involving nanoparticles and nanopores. In particular, the confinement of the ionic liquids in nanopores

leading to a change in the physicochemical properties of ionic liquids can be very important for preparation of supported ionic liquid membranes, where different types of ionic liquids are incorporated into nanopores.

Acknowledgements

P. Izak would like to thanks for financial support to the Czech Science Foundation (Grant No. 17-00089S) and V.V. Levdansky would like to thanks for financial support to the Ministry of Education, Youth and Sports of the Czech Republic (Grant No. LD14094).

References

- [1] J.-H. He, Y.-Q. Wan, L. Xu, Nano-effects, quantum-like properties in electrospun nanofibers, *Chaos, Solitons & Fractals* 33 (2007) 26–37.
- [2] J.-H. He, H.-Y. Kong, R.-R. Yang, H. Dou, N. Faraz, L. Wang, C. Feng, Review on fiber morphology obtained by bubble electrospinning and blown bubble spinning, *Thermal Science* 16 (2012) 1263–1279.
- [3] G. Guisbiers, Size-dependent materials properties toward a universal equation, *Nanoscale Res. Lett.* 5 (2010) 1132–1136.
- [4] Q. Jiang, C.C. Yang, Size effect in the phase stability of nanostructures, *Current Nanoscience* 4 (2008) 179–200.
- [5] W.H. Qi, Modeling the relaxed cohesive energy of metallic nanoclusters, *Materials Letters* 60 (2006) 1678–1681.
- [6] W.H. Qi, M.P. Wang, Size and shape dependent melting temperature of metallic nanoparticles, *Materials Chemistry and Physics* 88 (2004) 280–284.
- [7] C.C. Yang, S. Li, Investigation of cohesive energy effects on size-dependent physical and chemical properties of nanocrystals, *Phys. Rev. B* 75 (2007) 165413-1–165413-5.
- [8] S.C. Vanithakumari, K.K. Nanda, A universal relation for the cohesive energy of nanoparticles, *Phys. Lett. A* 372 (2008) 6930–6934.
- [9] C.Q. Sun, Size dependence of nanostructures: Impact of bond order deficiency, *Progress in Solid State Chemistry* 35 (2007) 1–159.
- [10] H.M. Lu, X.K. Meng, Theoretical model to calculate catalytic activation energies of platinum nanoparticles of different sizes and shapes, *J. Phys. Chem. C* 114 (2010) 1534–1538.

- [11] V.V. Levitskiy, J. Smolik, P. Moravec, Impurity trapping by aerosol particles, *Aerosol and Air Quality Research* 9 (2009) 257–265.
- [12] V.V. Levitskiy, J. Smolik, P. Moravec, Size effects in physicochemical transformations in aerosol systems with nanoparticles, *International Communications in Heat and Mass Transfer* 37 (2010) 593–595.
- [13] H.J. Xiao, H. Li, L. Chang, H.X. Zhang, H.C. Xuan, Theoretical prediction for the band gap of semiconductor nanoparticles as function of bond number, *Materials Chemistry and Physics* 184 (2016) 285–290.
- [14] C.X. Wang, Y.H. Yang, N.S. Xu, G.W. Yang, Thermodynamics of diamond nucleation on the nanoscale, *J. Am. Chem. Soc.* 126 (2004) 11303–11306.
- [15] N.M. Hwang, J.H. Hahn, D.Y. Yoon, Chemical potential of carbon in the low pressure synthesis of diamond, *J. Cryst. Growth* 160 (1996) 87–97.
- [16] C.Y. Zhang, C.X. Wang, Y.H. Yang, G.W. Yang, A nanoscaled thermodynamic approach in nucleation of CVD diamond on nondiamond surfaces. *J. Phys. Chem. B* 108 (2004) 2589–2593.
- [17] D. Kashchiev, G.M. van Rosmalen, Effect of pressure on nucleation in bulk solutions and solutions in pores and droplets, *Journal of Colloid and Interface Science* 169 (1995) 214–219.
- [18] A. Rai, K. Park, L. Zhou, M.R. Zachariah, Understanding the mechanism of aluminum nanoparticle oxidation, *Combustion Theory and Modelling* 10 (2006) 843–859.
- [19] V.M. Koshkin, V.V. Slezov, Doping nanoparticles, *Tech. Phys. Lett.* 30 (2004) 367–369.
- [20] G. Kaptay, The Gibbs equation versus the Kelvin and the Gibbs-Thomson equations to describe nucleation and equilibrium of nano-materials, *Journal of Nanoscience and Nanotechnology* 12 (2011) 1–9.
- [21] Q. Jiang, H.Y. Tong, D.T. Hsu, K. Okuyama, F.G. Shi, Thermal stability of crystalline thin films, *Thin Solid Films* 312 (1998) 357–361.
- [22] N.T. Gladkikh, A.P. Kryshtal, S.I. Bogatyrenko, Melting temperature of nanoparticles and the energy of vacancy formation in them, *Technical Physics* 55 (2010) 1657–1660.
- [23] Yu.I. Petrov, *Clusters and Small Particles*. Nauka, Moscow, 1986 (in Russian).
- [24] E.L. Nagaev, Small metal particles, *Sov. Phys. Usp.* 35 (1992) 747–782.
- [25] A.F. Lisovsky, On the application of Laplace pressure in the science of sintering, *Science of Sintering* 42 (2010) 357–362.

- [26] M. Shiraiwa, M. Ammann, T. Koop, U. Pöschl, Gas uptake and chemical aging of semisolid organic aerosol particles, *PNAS* 108 (2011) 11003–11008.
- [27] K.K. Nanda, S.N. Sahu, S.N. Behera, Liquid-drop model for the size-dependent melting of low-dimensional systems, *Phys. Rev. A* 66 (2002) 013208-1–013208-8.
- [28] G. Dudek, R. Turczyn, A. Strzelewicz, M. Krasowska, A. Rybak, Z.J. Grzywna, Studies of separation of vapours and gases through composite membranes with ferroferric oxide magnetic nanoparticles, *Sep. Purif. Technol.* 109 (2013) 55–63.
- [29] S.Sh. Rekhviashvili, E.V. Kishtikova, On the temperature of melting of nanoparticles and nanocrystalline substances, *Tech. Phys. Lett.* 32 (2006) 439–441.
- [30] M. Hirasawa, T. Orii, T. Seto, Size-dependent crystallization of Si nanoparticles, *Appl. Phys. Lett.* 88 (2006) 093119-1–093119-3.
- [31] D.B. Butrymowicz, J.R. Manning, M.E. Read, Diffusion in copper and copper alloys. Part 1. Volume and surface self-diffusion in copper, *J. Phys. Chem. Ref. Data* 2 (1973) 643–655.
- [32] F. Aqra, A. Ayyad, Surface energies of metals in both liquid and solid states, *Applied Surface Science* 257 (2011) 6372–6379.
- [33] Z.B. Wang, N.R. Tao, W.P. Tong, J. Lu, K. Lu, Diffusion of chromium in nanocrystalline iron produced by means of surface mechanical attrition treatment, *Acta Materialia* 51 (2003) 4319–4329.
- [34] J. Keizer, Diffusion effects on rapid bimolecular chemical reaction, *Chem. Rev.* 87 (1987) 167–180.
- [35] K.A. Connors, (1990) *Chemical Kinetics: The Study of Reaction Rates in Solution*, VCH, Weinheim/New York/Basel/Cambridge.
- [36] A.S. Iskovskikh, A.Ya. Katunin, I.I. Lukashevich, V.V. Sklyarevskii, V.V. Suraev, V.V. Filippov, N.I. Filippov, V.A. Shevtsov, Recombination and spin relaxation of hydrogen and deuterium atoms in molecular crystals, *Sov. Phys. JETP* 64 (1986) 1085–1094.
- [37] M.P. Singh, R.K. Singh, S. Chandra, Ionic liquids confined in porous matrices: Physicochemical properties and applications, *Progress in Materials Science* 64 (2014) 73–120.
- [38] L.A. Banu, D. Wang, R.E. Baltus, Effect of ionic liquid confinement on gas separation characteristics, *Energy Fuels*, 27 (2013) 4161–4166.

- [39] M.-A. Ne'ouze, J.L. Bideau, P. Gaveau, S. Bellayer, A. Vioux, Ionogels, new materials arising from the confinement of ionic liquids within silica-derived networks, *Chem. Mater.* 18 (2006) 3931–3936.
- [40] P. Liu, E. Harder, B.J. Berne, On the calculation of diffusion coefficients in confined fluids and interfaces with an application to the liquid-vapor interface of water, *J. Phys. Chem. B* 108 (2004) 6595–6602.
- [41] M. Attarian Shandiz, A. Safaei, S. Sanjabi, Z.H. Barber, Modeling the cohesive energy and melting point of nanoparticles by their average coordination number, *Solid State Communications* 145 (2008) 432–437.
- [42] L.N. Ho, Y. Schuurman, D. Farrusseng, B. Coasne, Solubility of gases in water confined in nanoporous materials: ZSM-5, MCM-41, and MIL-100, *J. Phys. Chem. C* 119 (2015) 21547–21554.
- [43] P. Li, Z. Wang, Z. Qiao, Y. Liu, X. Cao, W. Li, J. Wang, S. Wang, Recent developments in membranes for efficient hydrogen purification, *Journal of Membrane Science* 495 (2015) 130–168.

Captions

Fig. 1. Dependence of t_{mp}^* on the nanoparticle diameter d in diffusion of the copper-64 tracer in pure copper; solid line: the effect of the size-dependence of the cohesive energy and accordingly the activation energy for diffusion, dashed line: the Laplace pressure effect, dotted line: the joint influence of the mentioned effects.

Fig. 2. Dependence of k_{dp}^* on the nanoparticle diameter d ; solid lines: the effect of the size dependence of the cohesive energy and accordingly the activation energy for diffusion ($k_{dp}^* = k_{dp1}^*$), dashed lines: the Laplace pressure effect ($k_{dp}^* = k_{dp2}^*$); 1: $\alpha_1' = 2.5$; 2: $\alpha_1' = 1.5$; 3: $\alpha_2' = 0.2$; 4: $\alpha_2' = 0.5$.